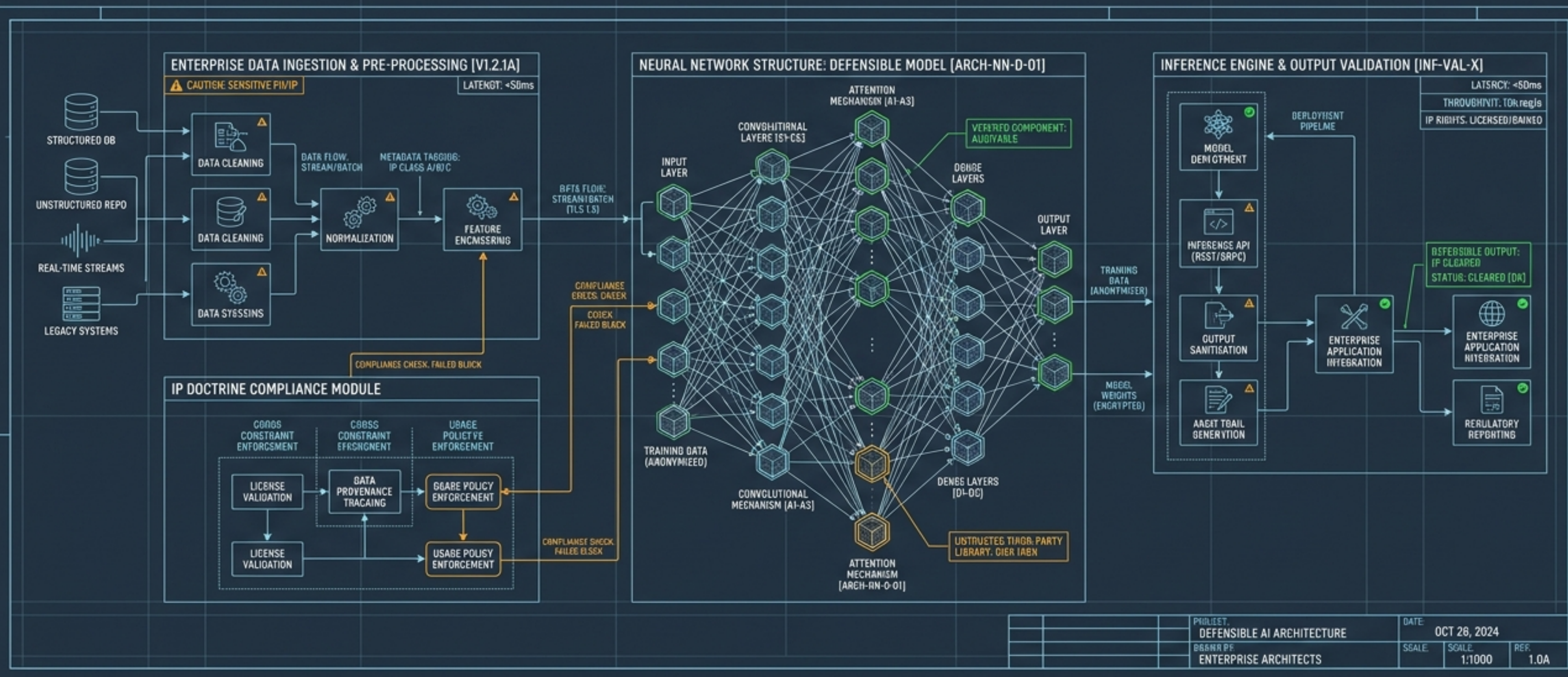


ARCHITECTING DEFENSIBLE AI SYSTEMS

TRANSLATING INTELLECTUAL PROPERTY DOCTRINE INTO SYSTEM DESIGN CONSTRAINTS



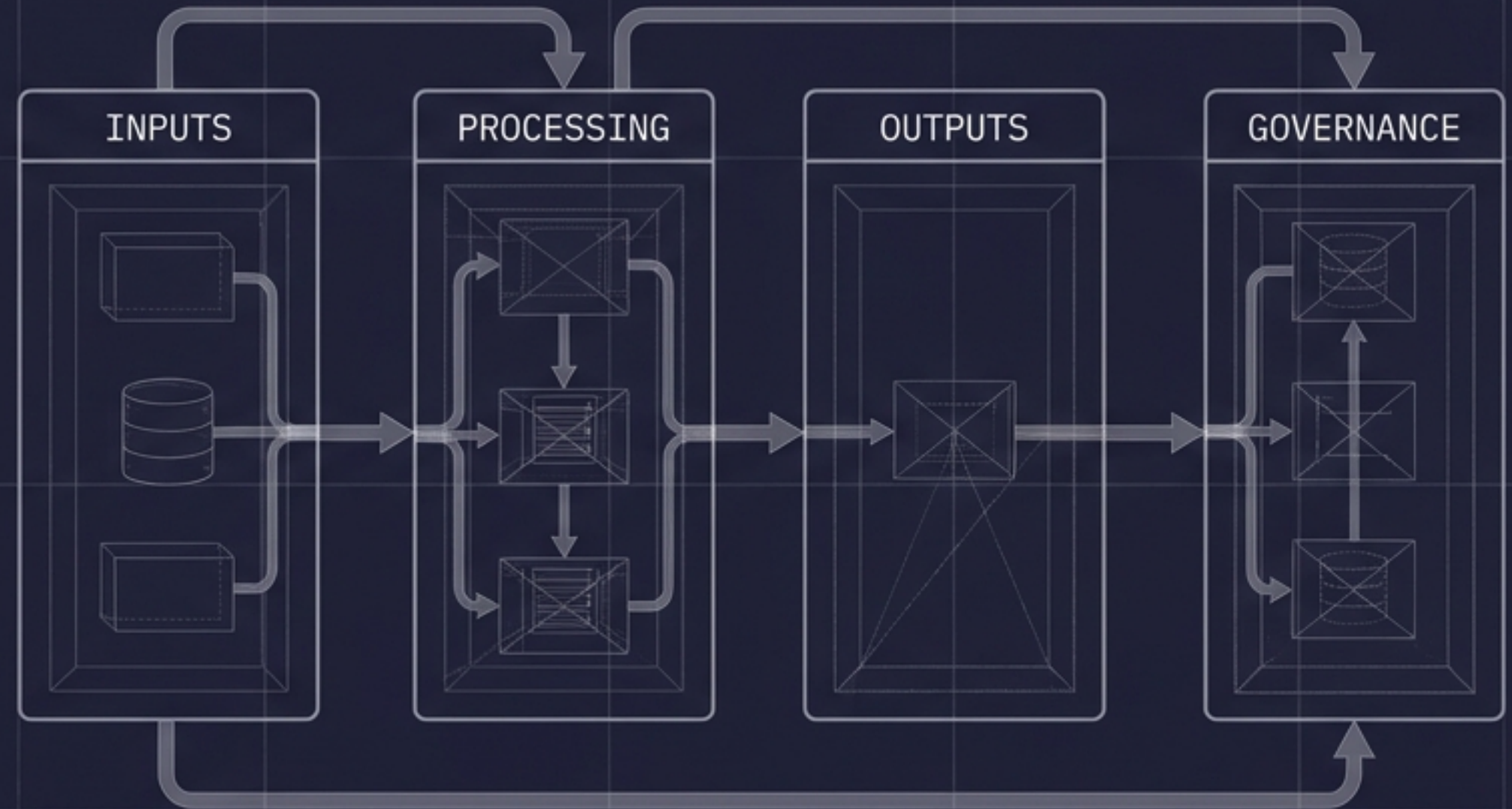


SYSTEM WARNING:
COMPLIANCE HAZARD DETECTED

System Constraints, Not Legal Advice

Unvetted training data and
unmonitored RAG pipelines
trigger systemic risk.

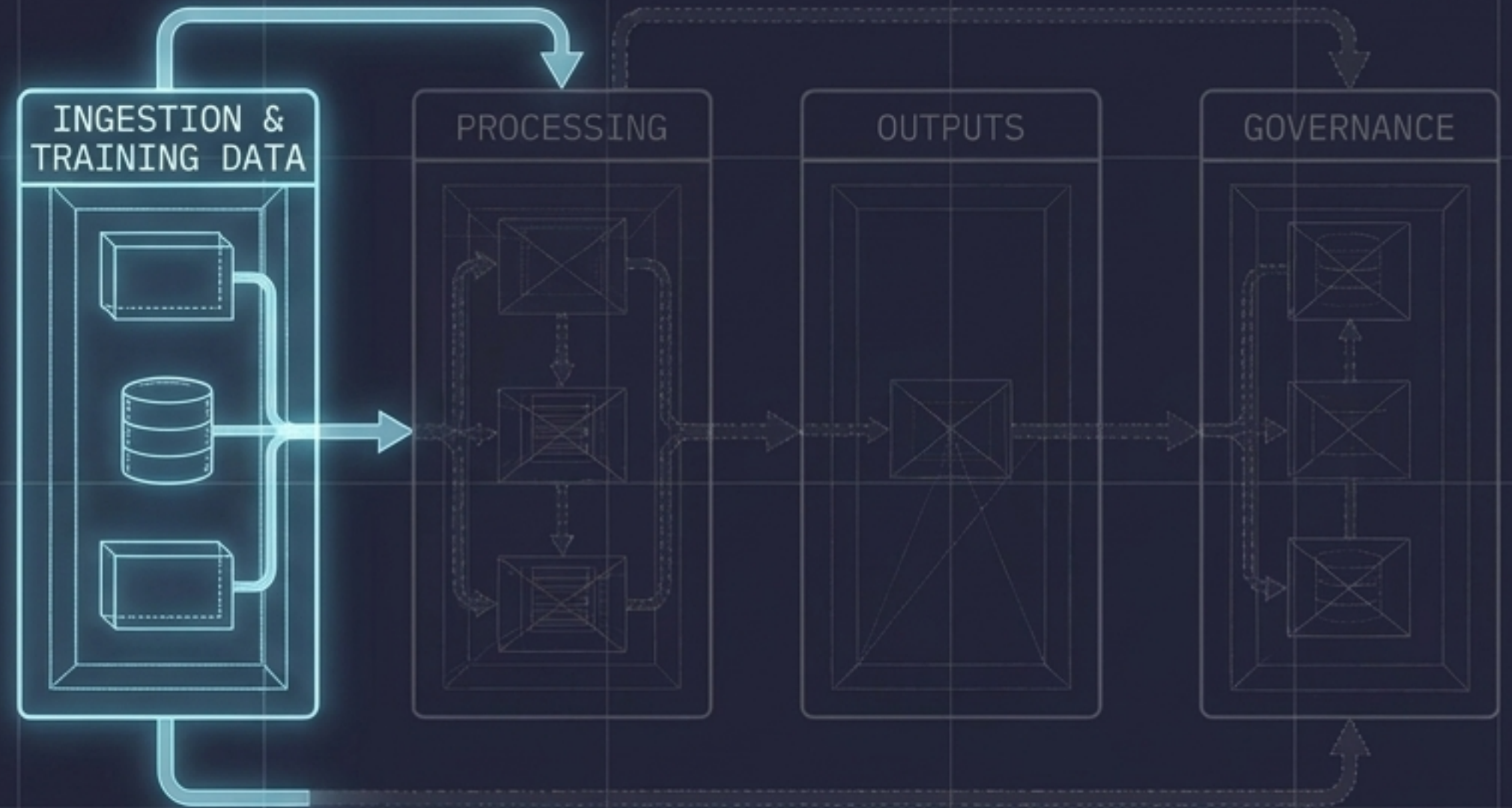
This session maps current IP
boundaries directly onto
enterprise system
architecture to design
compliant AI pipelines.



THE ARCHITECT'S QUESTION: How do legal boundaries dictate our technical system boundaries?

Phase I: The Ingestion Boundary

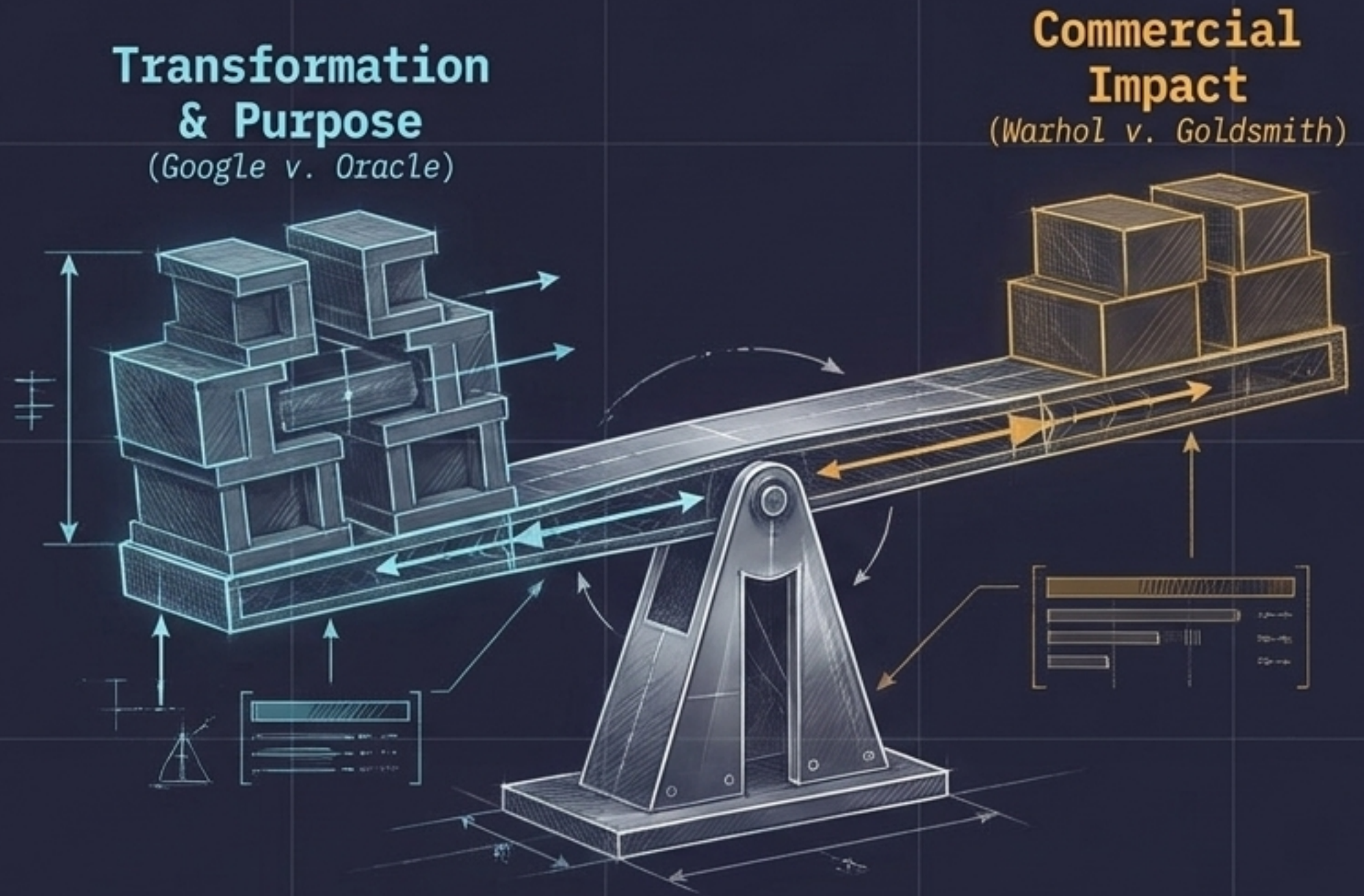
Securing the data pipeline against copyright infringement during training and retrieval.



THE ARCHITECT'S QUESTION: What is the origin and transformation state of our ingested data?

Fair Use is an Equilibrium, Not a Slogan

Warhol v. Goldsmith and *Google v. Oracle* establish that fair use is heavily dependent on the degree of technical transformation and the commercial purpose of the system.



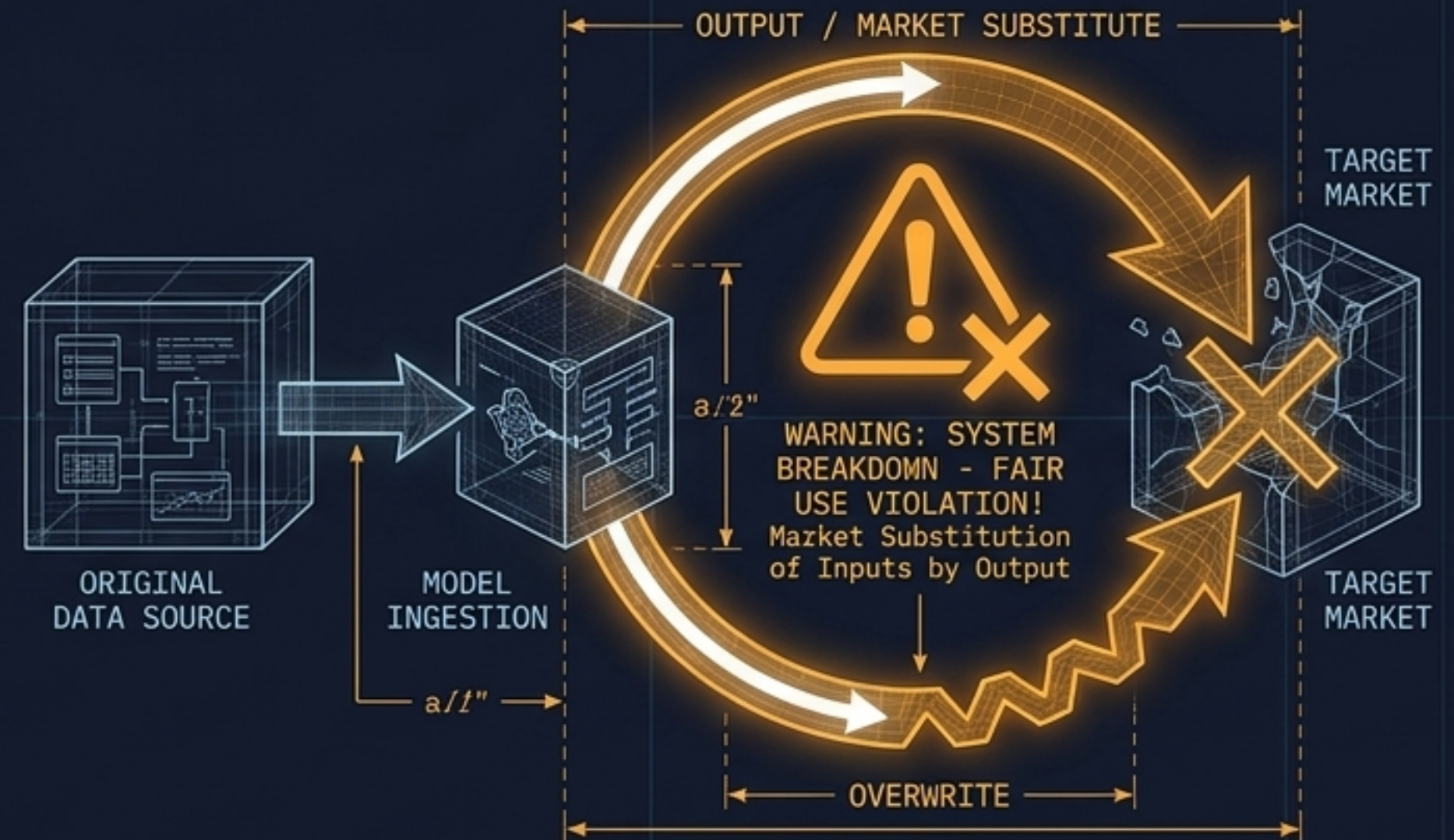
THE ARCHITECT'S QUESTION: Does our ingestion pipeline transform the data enough to shift the equilibrium?

The Gravity of Market Substitution

Case Map:

- Thomson Reuters v. Ross: Fair use rejected on headnotes because the system's output directly competed with the original market.
- Bartz and Kadrey: Some training mechanics treated as fair use, but the pirated acquisition pipeline remained legally exposed.

The Factor 4 Market Effect See-Saw



THE ARCHITECT'S QUESTION: Does our model's output act as a market substitute for its own training data?

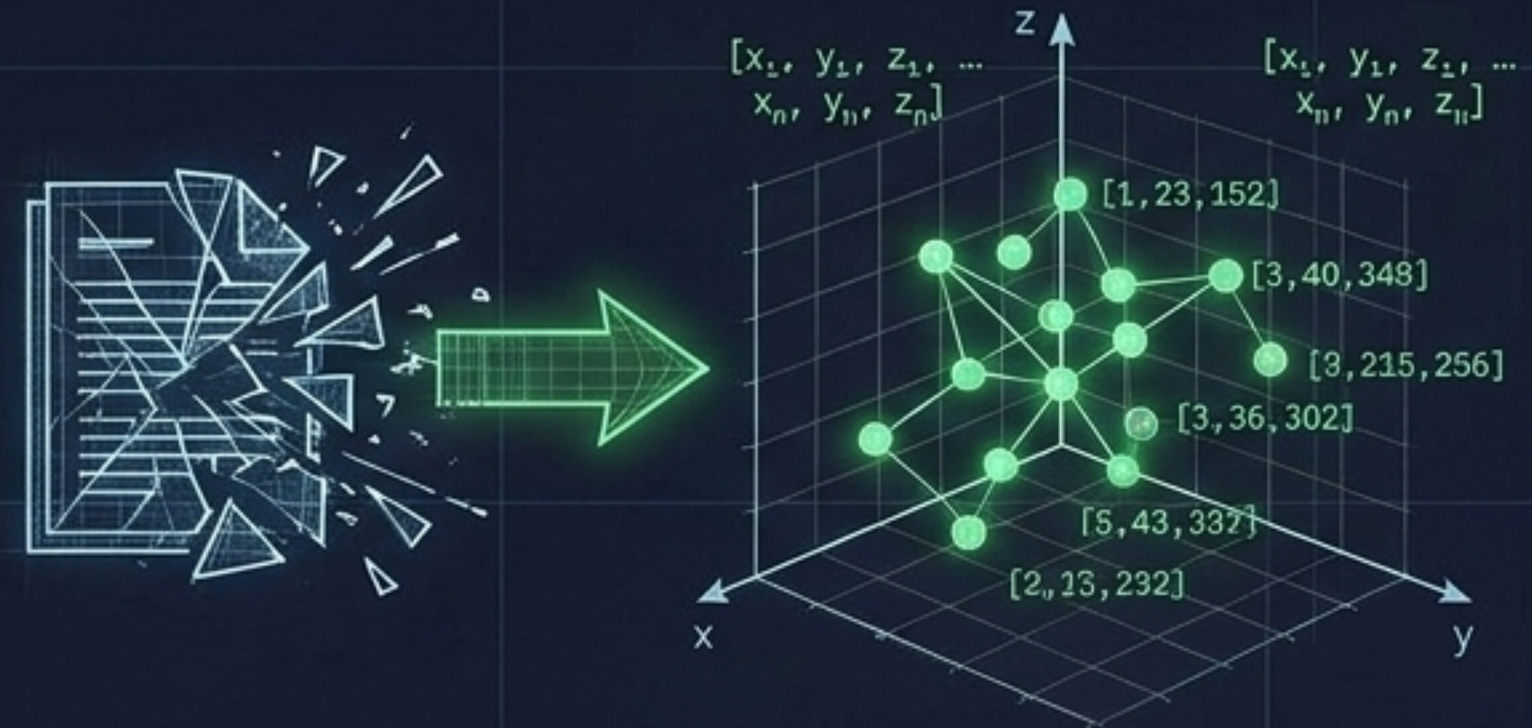
Architectural Divergence: RAG vs. Embeddings

Passing **verbatim** RAG passages directly into a **context window** carries fundamentally different reproduction risks than storing data as pure mathematical embeddings.

VERBATIM RAG



PURE VECTOR EMBEDDINGS



THE ARCHITECT'S QUESTION: Are we caching expressive text, or are we storing mathematical coordinates?

Phase II: The Core Architecture

Navigating the unsettled mechanics of model weights, trade secrets, and patent inventorship.

Master Blueprint: Intellectual Property Pipeline

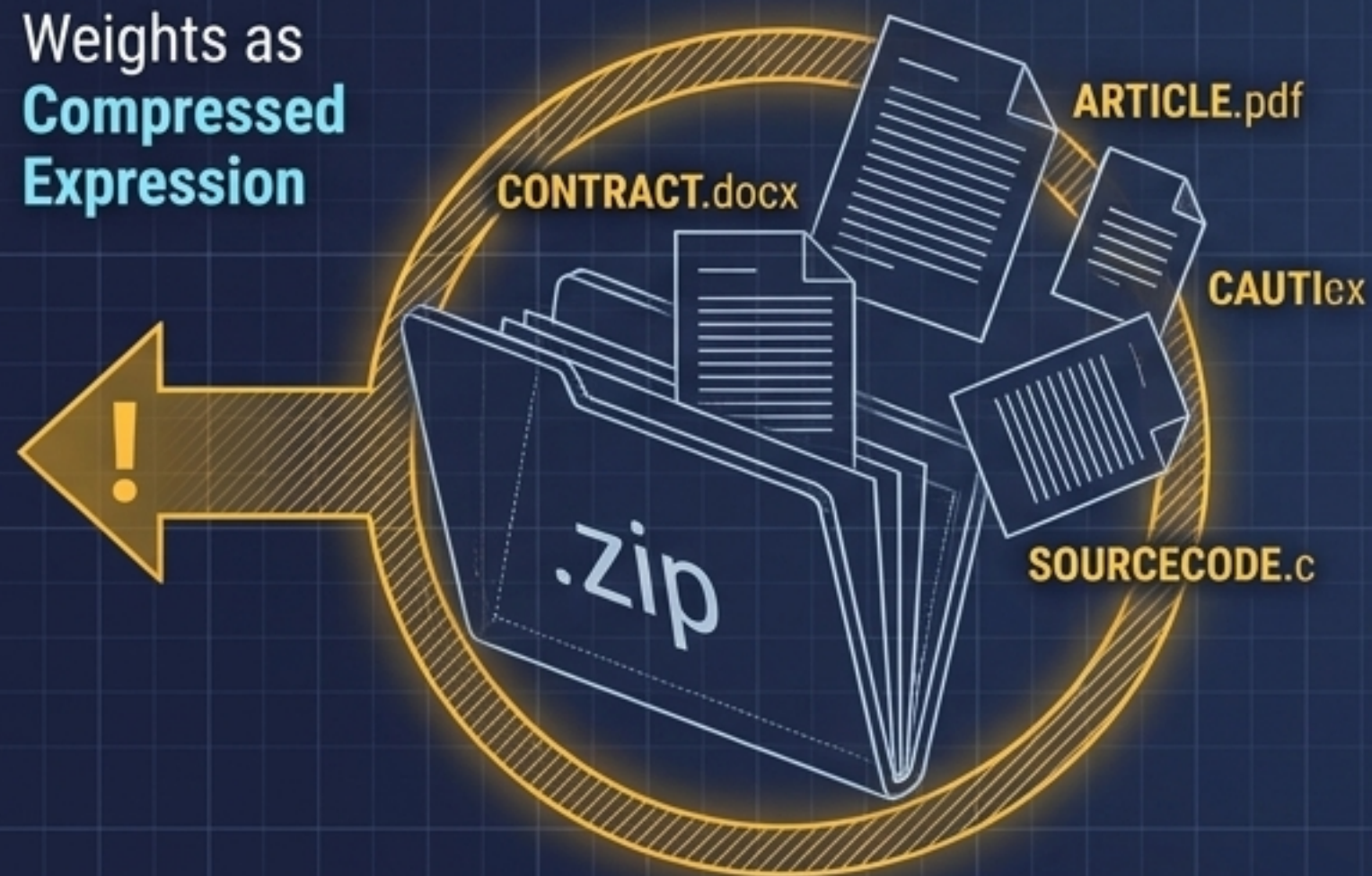


THE ARCHITECT'S QUESTION: How is intellectual property structurally embedded within the model?

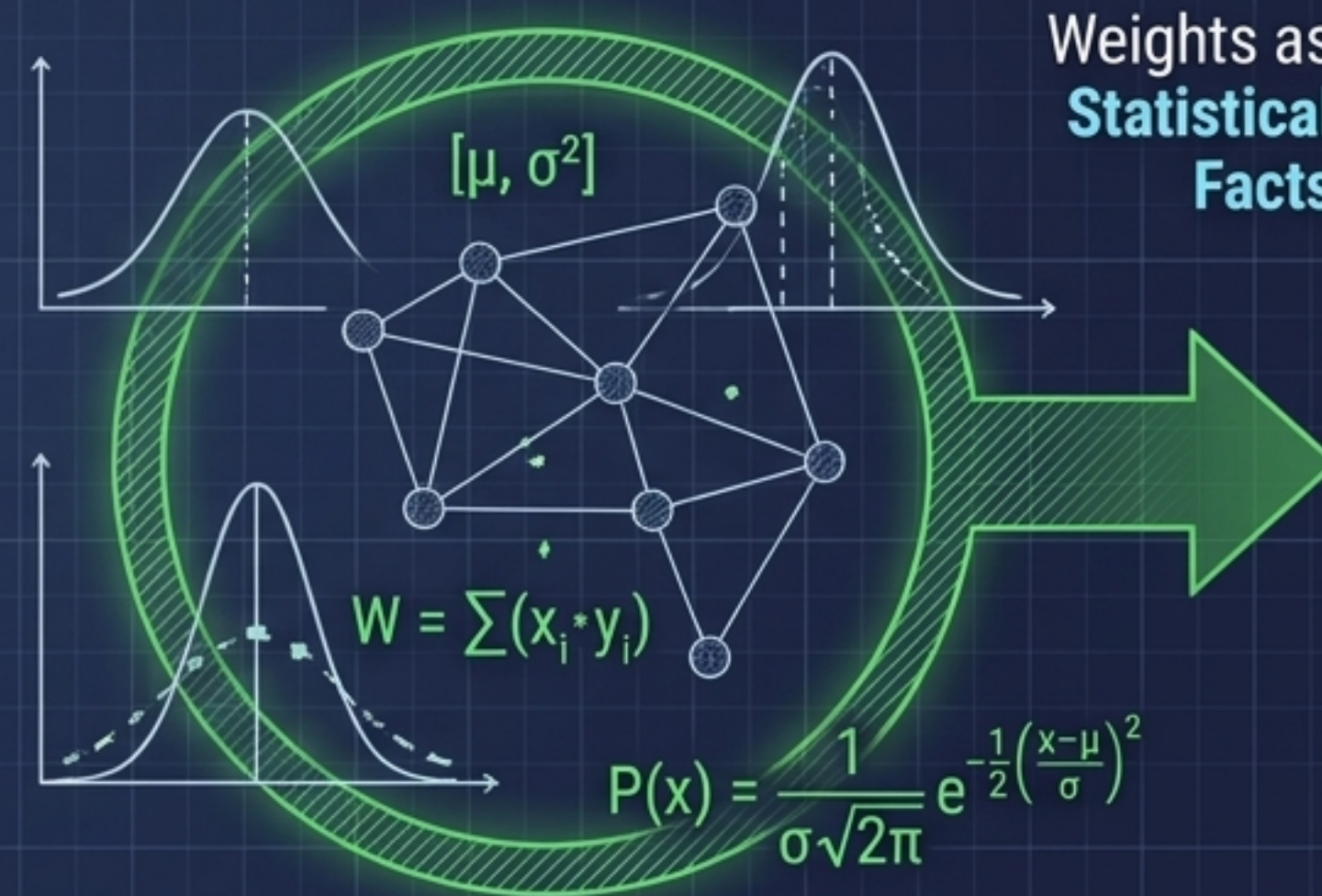
The Frontier (Rung 300): The Nature of Model Weights

Current scholarship is fractured. The legal system disagrees on whether model weights are a form of **compressed expression (high risk)** or **pure statistical facts (lower risk)**. **Acknowledge the disagreement; do not design around a fake holding.**

Weights as
**Compressed
Expression**



Weights as
**Statistical
Facts**

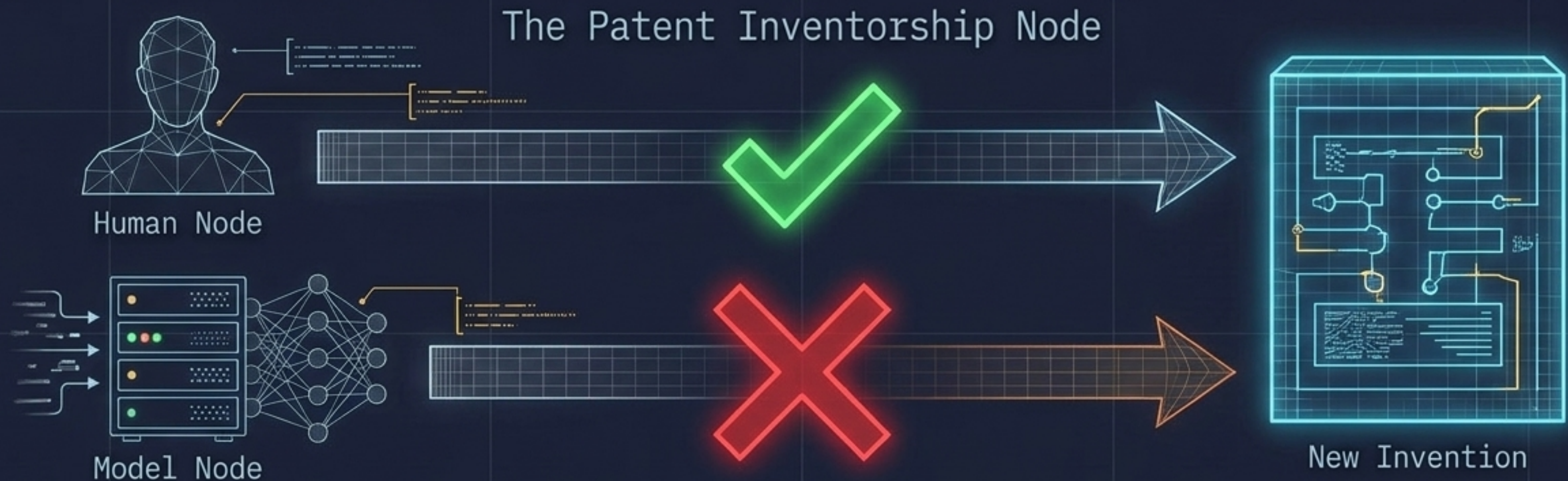


THE ARCHITECT'S QUESTION: Does our system architecture treat weights as protectable data or statistical exhaust?

Conceptions of Inventorship

Case Map: Thaler v. Vidal | Recentive v. Fox | USPTO Guidance

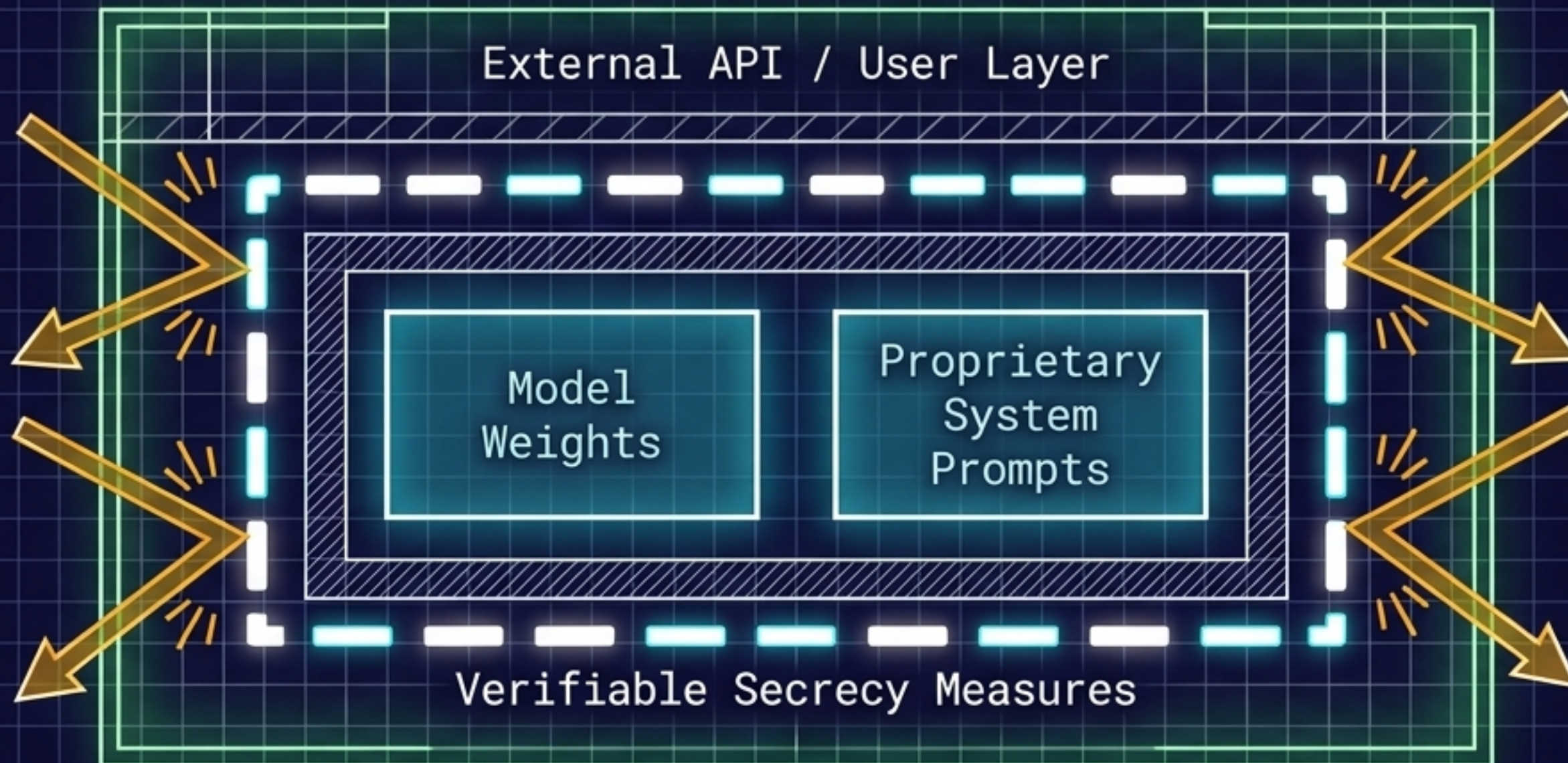
Key Takeaway: Only a natural person can conceive an invention. **Never list the model or the architecture as a co-inventor.**



THE ARCHITECT'S QUESTION: Which specific natural person conceived the output of this pipeline?

Technical Enclosures for Trade Secrets

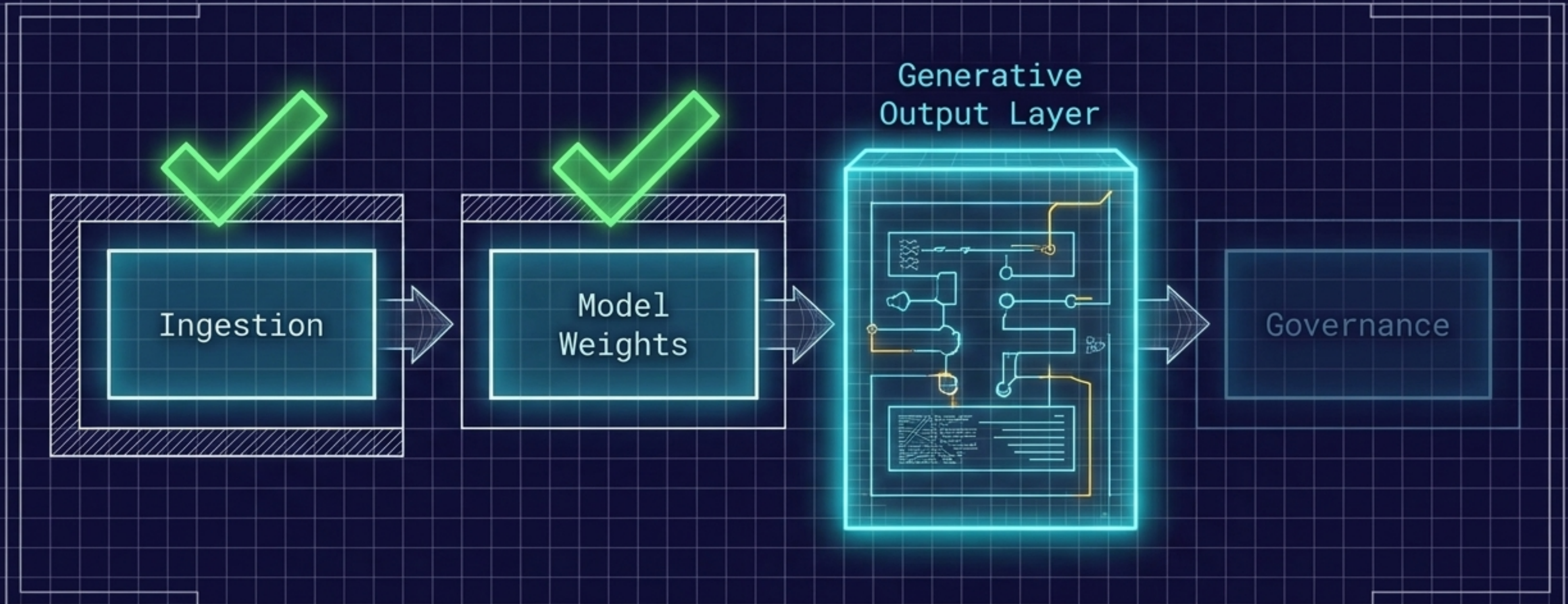
Key Takeaway: Legal protection for weights and proprietary system prompts evaporates without strict, verifiable technical secrecy measures and access controls at the deployment layer.



THE ARCHITECT'S QUESTION: What cryptographic or access barriers isolate our system prompts from user extraction?

Phase III: The Output Boundary

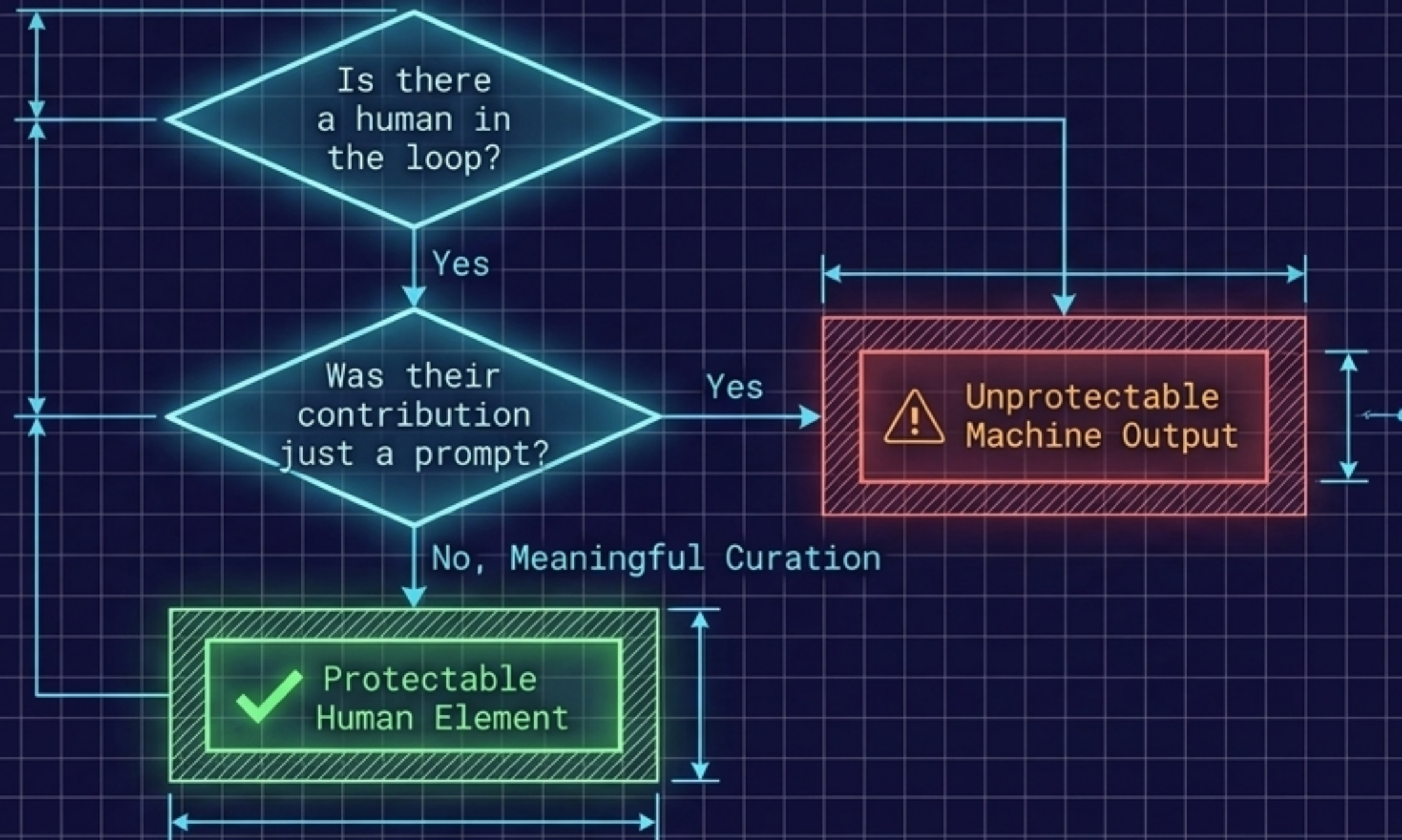
Determining human authorship and securing ownership of generated content.



THE ARCHITECT'S QUESTION: Who—or what—is mechanically generating the final output?

The Threshold of Human Authorship

Key Takeaway: Thaler v. Perlmutter establishes that prompting alone does not make the user an author. Pure machine output must be disclaimed.

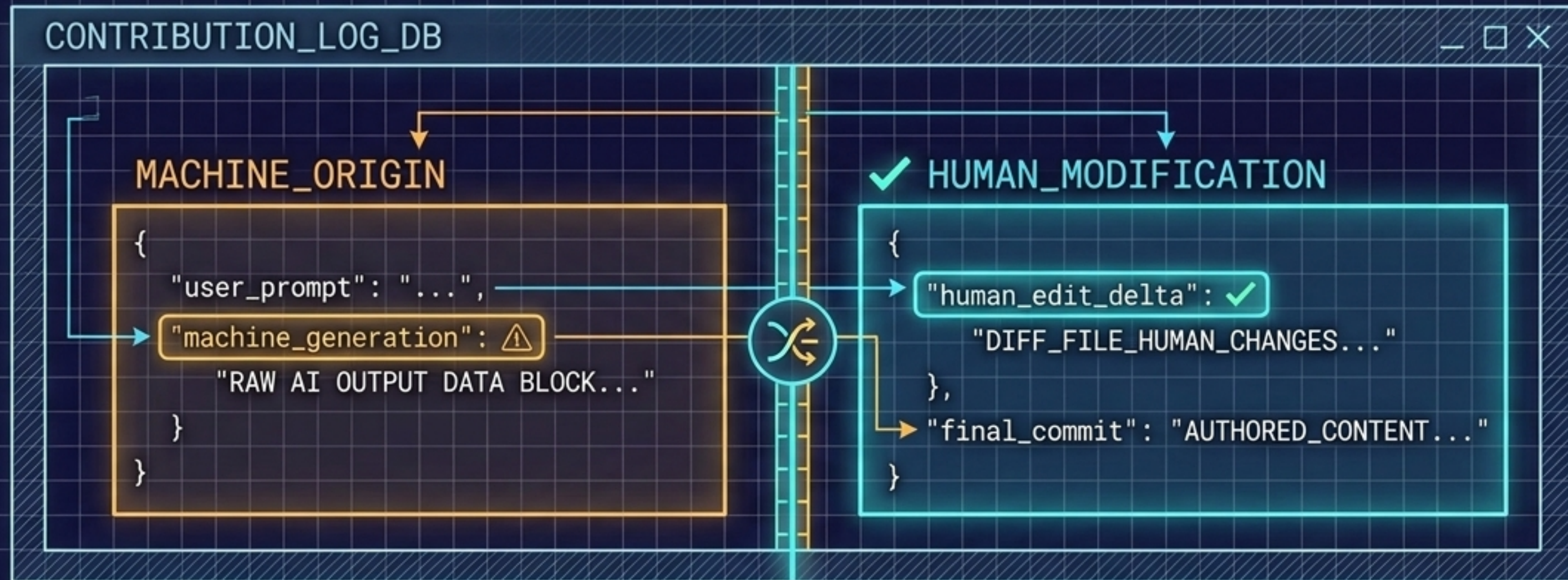


THE ARCHITECT'S QUESTION: Does our UI capture human curation or just trigger machine execution?

Systematizing the Contribution Log

Key Takeaway: To secure copyright on outputs, architects must design logging systems that explicitly track and isolate human edits from raw machine generation.

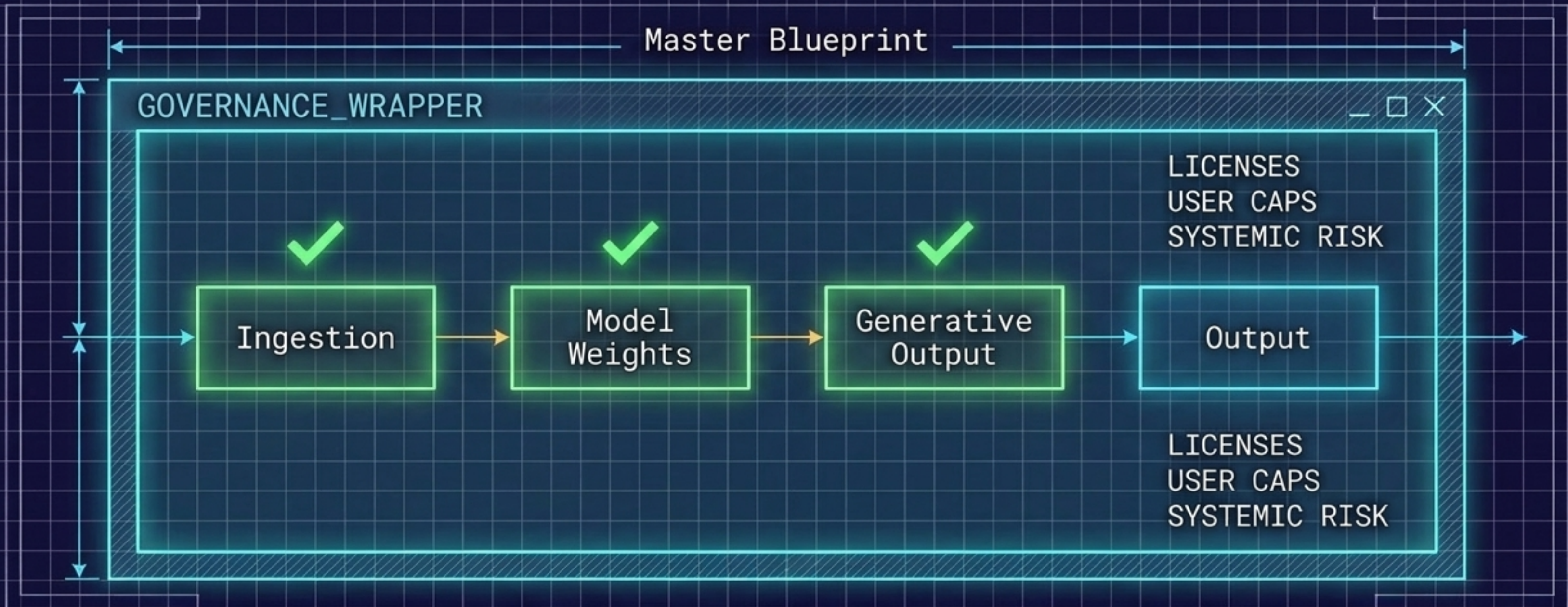
The Authorship Contribution Log



THE ARCHITECT'S QUESTION: How does our database schema delineate human modification from AI generation?

Phase IV: The Governance Wrapper

Auditing open-weights licenses, user caps, and systemic litigation risks.



THE ARCHITECT'S QUESTION: What external contractual limits bind our total system architecture?

The Open-Weights Illusion

Key Takeaway: Enterprise architects cannot treat modern model licenses like traditional open-source. Llama and Gemma utilize commercial user caps and behavioral use limits that require active system monitoring.

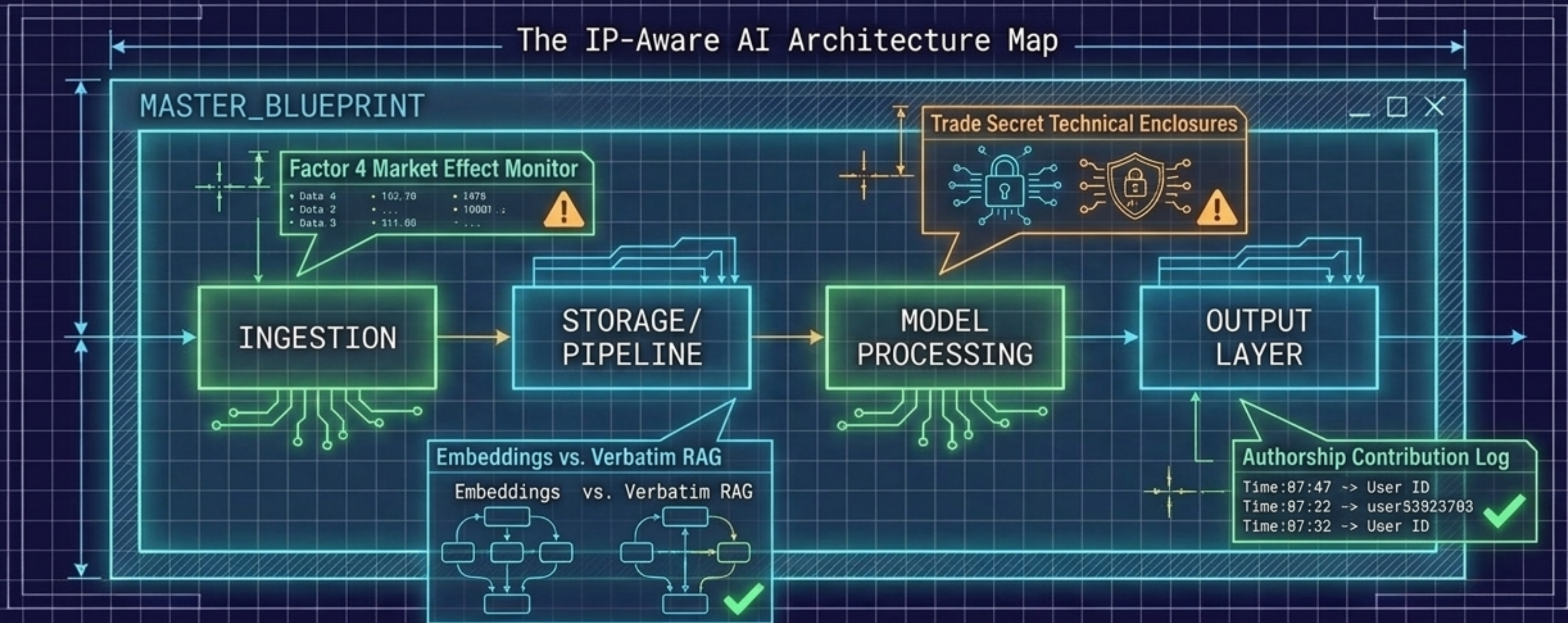
Open-Source vs. Open-Weights Governance Table

OPEN-SOURCE_SPECIFICATION	
OSAID 1.0 (Traditional Open-Source)	Llama/Gemma (Open-Weights)
 • Unrestricted Use: Modify, distribute, and commercialize without bounds. • Full Transparency: Complete access to source code, training data, and weights. • Community Governance: Developed and managed by open collaboration. • No User Caps: Unlimited scale and deployment.	 • Commercial User Caps (MAU Limits): Thresholds (e.g., >700M users) trigger different license terms. • Acceptable-Use Behavioral Limits: Restrictions on specific applications (e.g., critical infrastructure, high-risk). • Proprietary Data & Code: Weights are open, but training data and code are often closed. • Vendor Governance: Controlled by the model creator.

THE ARCHITECT'S QUESTION: Does our projected system scale trigger an open-weights commercial licensing threshold?

The Defensible Architecture Blueprint

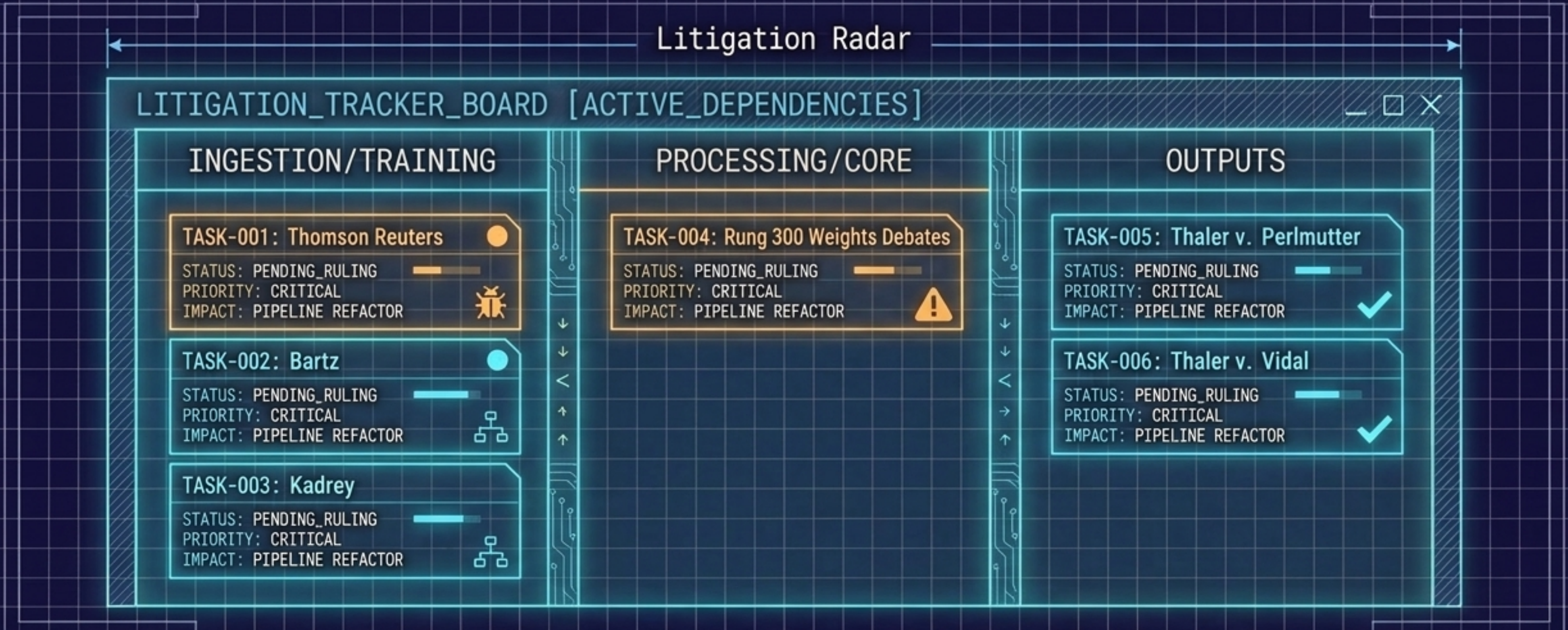
Key Takeaway: Legal compliance is not a post-launch review; it is a fundamental layer of technical system design controlled by the architect.



THE ARCHITECT'S QUESTION: Have we engineered out the IP risk at every discrete node of the pipeline?

Active Boundary Monitoring (Litigation Tracker)

Key Takeaway: System architectures must remain agile. As these specific cases resolve, they will trigger direct refactoring requirements within the corresponding pipeline components.



THE ARCHITECT'S QUESTION: Which component of our architecture is most vulnerable to tomorrow's ruling?